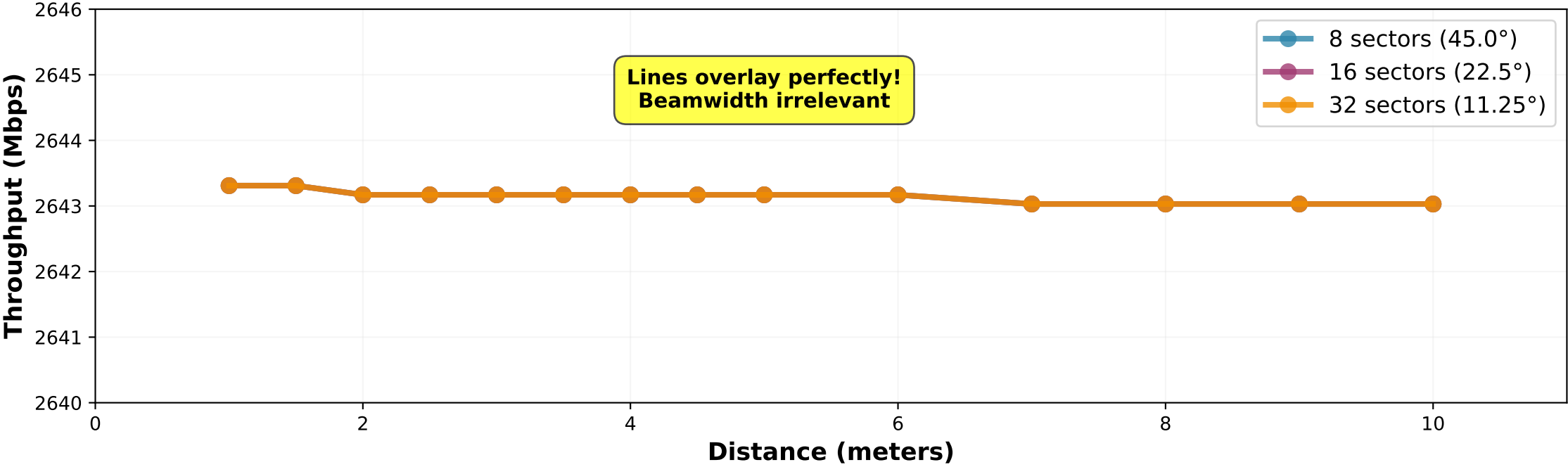
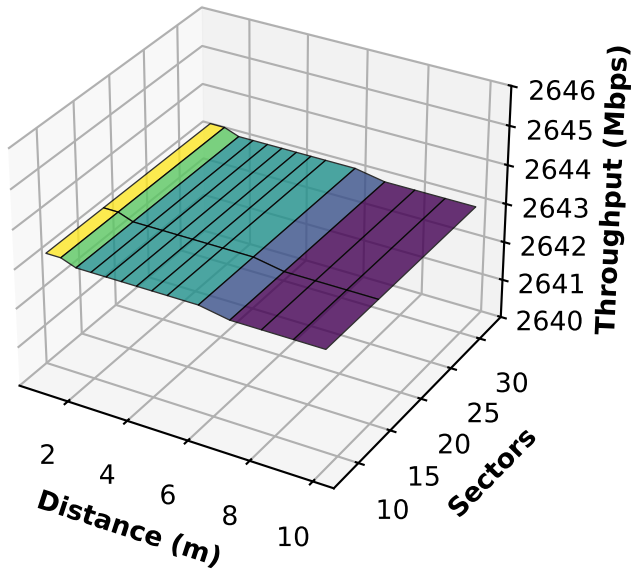


Complete Distance × Beamwidth Analysis - WiGig Performance Independence
MCS 12 | 60 GHz Data Center Network | 42 Test Combinations

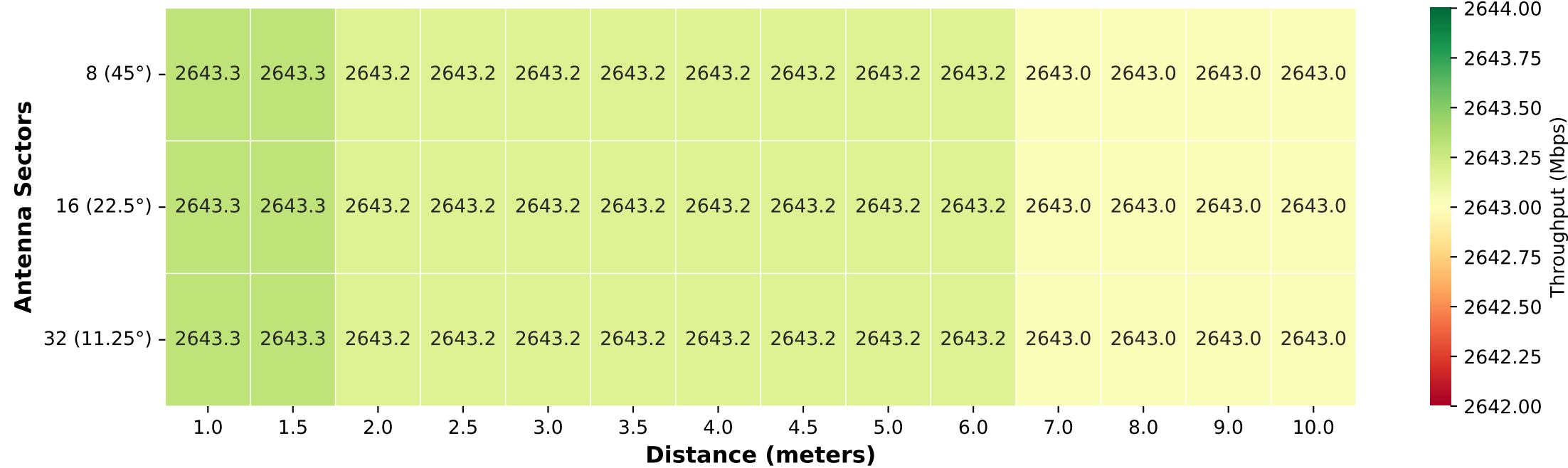
All Beamwidths Show Identical Performance



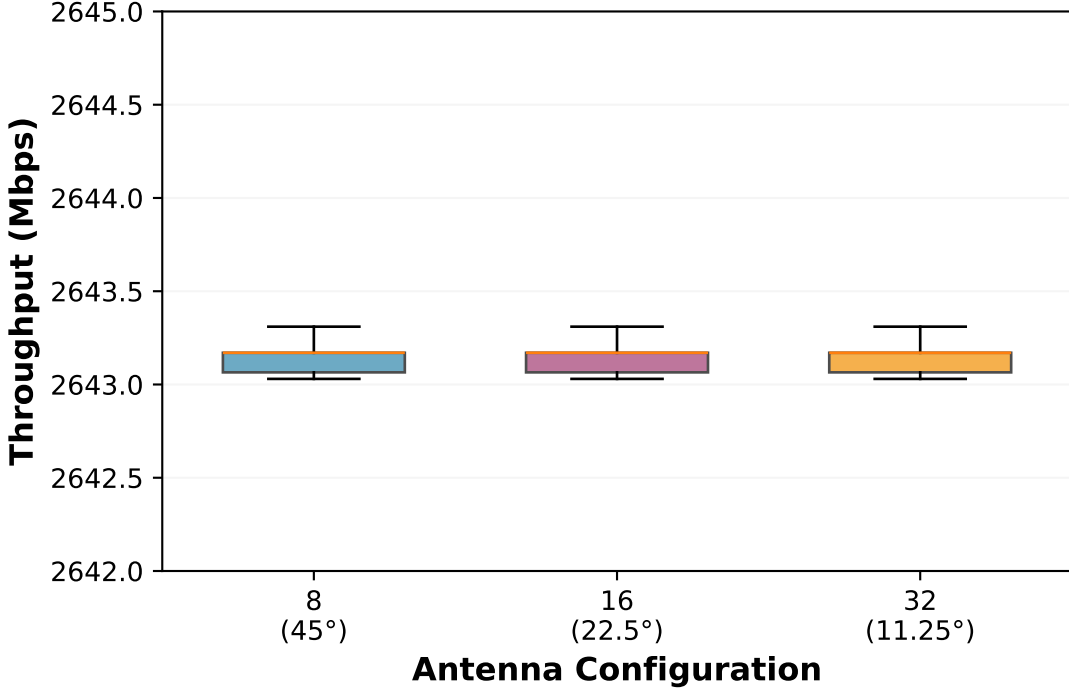
3D Surface: Flat Performance Plane



Heatmap: Uniform Performance Across All Parameters



Statistical Distribution by Beamwidth



COST-BENEFIT ANALYSIS

ANTENNA CONFIGURATIONS:

- 8 Sectors (45°) - Low cost
- 16 Sectors (22.5°) - Medium cost (~2x)
- 32 Sectors (11.25°) - High cost (~4x)

PERFORMANCE: ALL IDENTICAL

All: 2,643 Mbps (±0.01%)

RECOMMENDATION:

USE 8-SECTOR ANTENNAS

- Same performance as expensive options
- 50-75% cost savings
- Simpler installation
- No penalty at any distance

COST EXAMPLE (100 pairs):

8-sector: \$10k ✓ RECOMMENDED
16-sector: 20k ✗ Waste 10k
32-sector: 40k ✗ Waste 30k

KEY FINDINGS

PARAMETER INDEPENDENCE:

Distance: 1-10m
Variation: 0.01%

Beamwidth: 45°-11.25°
Variation: 0.00%

Interaction: None
Both irrelevant

TESTED:

42 combinations
ALL identical

WHY IT WORKS:

- Beamforming
- High gain
- LOS propagation

LIMITATION:

Requires LOS only

STATUS: COMPLETE